Self-Check Questions: An Introduction to Real Analysis

Chapters 1–6

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How to use these questions. Answer without consulting the notes, then check against the cited section. Give complete hypotheses when stating a result. Justify examples and counterexamples. Proof questions concentrate on the arguments developed in the chapter; other results are primarily tested through statements and applications. No solutions are included.

Unless specified otherwise, all sets are subsets of $\mathbb{R}$, and openness, closedness, limits, and compactness are taken in $\mathbb{R}$. Use $\mathbb{N} = \{1, 2, 3, \ldots\}$.

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1 The Real Numbers

1.1 Completeness of $\mathbb{R}$

Source: §1.1, pp. 1–3.

Suprema and infima

1. Define an upper bound and a lower bound of a set A. What does it mean for A to be bounded above, bounded below, or bounded? State an equivalent definition of boundedness using absolute values.
2. Give a set that is bounded below but unbounded above. Identify all its lower bounds and explain why it has no upper bound.
3. Define the supremum and infimum of a set. In each definition, distinguish the bound condition from the leastness or greatestness condition.
4. Let $A \subseteq \mathbb{R}$ be nonempty and let $s \in \mathbb{R}$ be an upper bound of A. State the ϵ-approximation property equivalent to $s = \sup A$, with all quantifiers. State the analogous characterization of $m = \inf A$ for a lower bound $m \in \mathbb{R}$.
5. Define the maximum and minimum of a set. How do they differ from the supremum and infimum?
6. What can be said about the boundedness, supremum, infimum, maximum, and minimum of a nonempty finite set of real numbers?
7. Find the supremum and infimum of $A = \{1/n : n \in \mathbb{N}\}$, and decide whether it has a maximum and a minimum. Verify the least-upper-bound and greatest-lower-bound conditions.
8. Give an example of each of the following: a bounded nonempty set with neither a maximum nor a minimum; one with both; and one with a maximum but no minimum. Find the supremum and infimum of each.
9. State the least-upper-bound formulation of completeness of $\mathbb{R}$ (Theorem 1.7), including all hypotheses. Why can neither nonemptiness nor boundedness above be omitted?
10. For $-A = \{-x : x \in A\}$, explain the relationship between $\inf A$ and $\sup(-A)$, including when these real numbers exist. Deduce the greatest-lower-bound version of completeness.
11. Give a nonempty subset of $\mathbb{Q}$ that is bounded above by a rational number but has no supremum in $\mathbb{Q}$. Explain why no rational upper bound is least. Does the same set have a supremum in $\mathbb{R}$?

Cauchy sequences

12. State the definition of a Cauchy sequence, with the quantifiers in order. How does it differ from the definition of convergence to a specified limit?
13. State the Cauchy characterization of convergence for real sequences (Theorem 1.10). Which implication expresses completeness?
14. Give a Cauchy sequence of rational numbers that has no limit in $\mathbb{Q}$. Explain why it is Cauchy and why it cannot converge to a rational number. What happens when it is regarded as a sequence in $\mathbb{R}$?

1.2 Open sets

Source: §1.2, pp. 3–5.

15. State the definition of an open subset of $\mathbb{R}$ using intervals of radius δ. May δ depend on the point?
16. Define an interior point and a boundary point of a set. Must either kind of point belong to the set?
17. Characterize open sets in terms of interior points.
18. Given $x \in (a, b)$, with $a < b$ finite, choose an explicit $\delta > 0$ such that $(x - \delta, x + \delta) \subseteq (a, b)$. How would you choose δ for $x \in (a, \infty)$ or $x \in (-\infty, b)$?
19. Find all interior and boundary points of each of $(0, 1)$, $[0, 1]$, and $(0, 1]$.
20. State the closure properties of open sets under unions and intersections. Specify which collections may be arbitrary and which must be finite.
21. Give a countable family of open intervals whose intersection is not open. Compute the intersection and explain why it is not open.
22. State the interval decomposition result for open subsets of $\mathbb{R}$. What can be required of the intervals and their number?
23. What does it mean for a property to be a topological property? Name two examples mentioned in the notes.
24. Define a neighborhood of x and a δ-neighborhood of x, using the convention in these notes. Must a neighborhood be open? Must it contain x?
25. Give a neighborhood of a point that is not an open set. For $a < b$, determine exactly which points of $[a, b]$ have $[a, b]$ as a neighborhood.
26. Restate the definition of an open set in terms of neighborhoods, and explain why it agrees with the interval definition.
27. Define what it means for $B \subseteq A$ to be relatively open in A. Which set in the definition must be open in $\mathbb{R}$?
28. Give two subsets of $[0, 1]$ that are open in $[0, 1]$ but not open in $\mathbb{R}$. For each, exhibit an open set $U \subseteq \mathbb{R}$ whose intersection with $[0, 1]$ is the given subset.
29. Define a relative neighborhood of $x \in A$. Express the definition using a neighborhood in $\mathbb{R}$.
30. State the relative-neighborhood characterization of relative openness (Proposition 1.22).
31. Prove Proposition 1.22 in both directions. In the direction that starts with a relative neighborhood at each point, construct an open set $U \subseteq \mathbb{R}$ such that $B = A \cap U$. Why is it useful to replace the given neighborhoods by open neighborhoods?

1.3 Closed sets

Source: §1.3, pp. 5–6.

32. Define a closed subset of $\mathbb{R}$ using its complement. In which ambient set is the complement taken?
33. Define a sequentially closed subset of $\mathbb{R}$. Where must the sequence lie, and where is its convergence considered?

34. State the relationship between closedness and sequential closedness for subsets of $\mathbb{R}$.

35. Verify that $[0, 1]$ is closed in two ways: using its complement and using convergent sequences. Which property of limits is needed for the second argument?

36. State the closure properties of closed sets under unions and intersections, specifying the permitted sizes of the collections. Explain how they follow from the corresponding facts about open sets.

37. Give a countable family of closed intervals whose union is not closed. Compute the union and explain why it is not closed.

38. Which subsets of $\mathbb{R}$ are both open and closed? Can a subset of $\mathbb{R}$ be neither open nor closed?

39. Give a set that is neither open nor closed. Show the failure of openness using neighborhoods and the failure of closedness using a convergent sequence.

1.4 Accumulation points and isolated points

Source: §1.4, pp. 6–7.

40. Define an accumulation point of a set using intervals. What other names are used for an accumulation point? Why is the requirement that the nearby point differ from the given point essential?

41. Why must every neighborhood of an accumulation point of A contain infinitely many points of A, rather than just one other point?

42. State the sequential characterization of an accumulation point. What restriction on the terms of the sequence must be included? Must those terms be pairwise distinct in this characterization?

43. Must every boundary point of a set be an accumulation point? Must every accumulation point be a boundary point? Give a simple counterexample to each implication that fails.

44. Explain how to pass from the interval definition of an accumulation point to the sequential characterization and back.

45. State the characterization of closed sets in terms of their accumulation points.

46. Give an infinite subset of $\mathbb{R}$ with no accumulation points in $\mathbb{R}$. Explain why it has none.

47. Find all accumulation points of $A = \{1/n : n \in \mathbb{N}\}$. Which of them belong to A? Explain why there are no others.

48. For $a < b$, find all accumulation points of (a, b) and of $[a, b]$. Which accumulation points belong to each set?

49. For $a < c < b$, find all accumulation points of $(a, c) \cup (c, b)$. What effect does removing c have on the set of accumulation points?

50. Define an isolated point of a set. Must it belong to the set? State the dichotomy between isolated and accumulation points for points that belong to a set. Does the same dichotomy hold for every point of $\mathbb{R}$?

51. Show that every point of $\{1/n : n \in \mathbb{N}\}$ is isolated. For each n, give an explicit positive radius that isolates $1/n$, treating $n = 1$ separately if necessary.

52. Can an interval have an isolated point? Distinguish intervals with more than one point from singleton intervals.

1.5 Compact sets

Source: §1.5, pp. 7–10.

Sequential compactness

53. Define sequential compactness. Which sequences are quantified over, and where must the subsequential limit belong?

54. Give a bounded set that is not sequentially compact. Exhibit a sequence in the set with no subsequence converging to a point of the set. Check that every term belongs to the set.

55. Give a closed set that is not sequentially compact, and exhibit a sequence that demonstrates this failure.

56. State the closed-and-bounded characterization of sequential compactness in $\mathbb{R}$ (Theorem 1.40).

57. Prove that a sequentially compact set is closed by considering a convergent sequence in the set. Why must the limit of a convergent subsequence agree with the limit of the original sequence?

58. Prove that a sequentially compact set is bounded. If the set were unbounded, how would you choose a sequence that rules out every convergent subsequence?

59. State the Bolzano–Weierstrass theorem for real sequences. Use it to prove that a closed, bounded subset of $\mathbb{R}$ is sequentially compact. Where is each hypothesis used?

60. State the maximum-and-minimum result for sequentially compact subsets of $\mathbb{R}$ (Proposition 1.41), including the necessary nonemptiness assumption. Why is that assumption needed?

61. Prove that a nonempty sequentially compact set contains its supremum and infimum. Explain how the definition of supremum gives a suitable sequence, and where completeness and closedness enter.

Open covers and compactness

62. Define an open cover of a set and a finite subcover of a given open cover. Must the sets in a cover be contained in the set being covered? Must a subcover use sets from the original cover?

63. Define compactness using open covers. Is it enough to find one open cover that has a finite subcover?

64. Construct an open cover of $\mathbb{N}$ with no finite subcover. Explain why an arbitrary finite subcollection fails to cover $\mathbb{N}$.

65. Use the cover $\{(-n, n) : n \in \mathbb{N}\}$ to prove directly from the open-cover definition that every compact subset of $\mathbb{R}$ is bounded.

66. Construct an open cover of $(0, 1)$ with no finite subcover. Verify that it covers every point and that any finite subcollection misses some points near an endpoint.

67. For integers $n \geq 0$, let

$$I_n = \left(2^{-n} - 2^{-(n+1)},\ 2^{-n} + 2^{-(n+1)}\right).$$

Find $\bigcup_{n=0}^{\infty} I_n$ and $\bigcup_{n=0}^{N} I_n$. Why does this family cover $(0, 1)$ but have no finite subcover of $(0, 1)$?

68. For the same intervals I_n, add $J = (-\delta, \delta)$, where $\delta > 0$. Show that this gives an open cover of $[0,1]$ with a finite subcover by specifying a sufficient condition on N for $J, I_0, \dots, I_N$ to cover $[0,1]$. Why does verifying this particular cover alone not prove compactness?

69. State the closed-and-bounded characterization of compact subsets of $\mathbb{R}$ (Theorem 1.46) and the interval version of the Heine–Borel theorem quoted in the notes.

70. What is the relationship between compactness and sequential compactness for subsets of $\mathbb{R}$? Explain how Theorems 1.40 and 1.46 establish it.

71. Decide which of the following sets are compact, and justify each answer:

$$\varnothing, \quad \{2\}, \quad [0,1], \quad (0,1], \quad \mathbb{N}, \quad \{1/n : n \in \mathbb{N}\}, \quad \{0\} \cup \{1/n : n \in \mathbb{N}\}.$$

2 Limits of Functions

2.1 Limits

Source: §2.1, pp. 11–14.

1. State the ϵ-δ definition of $\lim_{x\to c} f(x) = L$ for $f : A \to \mathbb{R}$. Include the hypothesis on c, the domain restriction, and the order of all quantifiers.
2. Must c belong to the domain for $\lim_{x\to c} f(x)$ to exist? If $f(c)$ is defined, does changing that value change the limit? Explain using the definition.
3. Write the precise negation of $\lim_{x\to c} f(x) = L$, for a specified $L \in \mathbb{R}$. How does this differ from saying that f has no finite limit at c?
4. Express convergence of $f(x)$ to L in terms of a limit of $|f(x)-L|$, and explain the equivalence directly from the definitions.
5. For $A = [0, \infty) \setminus \{9\}$, find
$$\lim_{x\to 9} \frac{x-9}{\sqrt{x}-3}.$$
Prove your answer from the definition by giving an explicit δ for each $\epsilon > 0$.
6. Define a punctured (or deleted) neighborhood of c and a punctured δ-neighborhood of c. Must such a neighborhood contain c? Must it be an open set?
7. State the neighborhood formulation of $\lim_{x\to c} f(x) = L$. Explain its equivalence with the ϵ-δ formulation in both directions.
8. State the sequential characterization of a function limit (Theorem 2.4). Which sequences must be considered, and why are terms equal to c excluded?
9. Prove that $\lim_{x\to c} f(x) = L$ implies the sequential condition in Theorem 2.4. Explain how the choices of δ and then N fit together.
10. Prove the converse in Theorem 2.4 by contraposition. Starting from failure of convergence to the proposed L, construct a sequence $x_n \to c$ for which $f(x_n) \not\to L$. Why must the same positive error threshold work for every n?
11. State the two sequential criteria in Corollary 2.5 for showing that a finite function limit does not exist. Is checking one sequence whose function values converge sufficient to prove that a function limit exists?
12. Define the sign function sgn on $\mathbb{R}$. Use two sequences to show that it has no limit at zero.
13. Show, using the sequential criterion, that $1/x$ has no finite limit as $x \to 0$. Can a single sequence establish this?
14. Show that $\sin(1/x)$ has no limit as $x \to 0$ by constructing two explicit sequences tending to zero whose function values have different limits.

2.2 Left, right, and infinite limits

Source: §2.2, pp. 14–16.

15. State the definitions of the right-hand and left-hand limits of $f : A \to \mathbb{R}$ at c. What accumulation condition on A is needed for each side? What would go wrong if the relevant side contained no domain points sufficiently close to c?

16. Find the two one-sided limits of $\operatorname{sgn} x$ at zero and verify them from the definitions.

17. Define $\lim_{x\to\infty} f(x) = L$ and $\lim_{x\to-\infty} f(x) = L$. Include the necessary assumptions on the domain and the quantifiers.

18. How can a limit as $x \to \infty$ or $x \to -\infty$ be rewritten using the substitution $x = 1/t$? Specify the side from which t approaches zero and how the domain is transformed.

19. Find and prove the limits of $x/\sqrt{1+x^2}$ as $x \to \infty$ and as $x \to -\infty$. Why must the identity $\sqrt{x^2} = |x|$ be handled carefully?

20. State precisely what it means for $f(x)$ to diverge to $+\infty$ or $-\infty$ as $x \to c$. Do these statements assert the existence of a real-valued limit?

21. Prove directly from the appropriate definitions that

$$\lim_{x\to 0} \frac{1}{x^2} = +\infty, \qquad \lim_{x\to\infty} \frac{1}{x^2} = 0.$$

Give explicit choices of the required radius or cutoff.

22. Write the quantified definitions of $\lim_{x\to 0^+} f(x) = +\infty$ and $\lim_{x\to 0^-} f(x) = -\infty$. Apply them to $f(x) = 1/x$.

23. Why does $1/x$ tend to neither $+\infty$ nor $-\infty$ as $x \to 0$, even though it is unbounded in every punctured neighborhood of zero?

24. Show that $x^{-1}\sin(1/x)$ has neither a finite limit nor divergence to either sign of infinity as $x \to 0^+$ or $x \to 0^-$. Construct sequences that demonstrate its behavior. What follows for the two-sided limit?

25. Define $\lim_{x\to\infty} f(x) = -\infty$ and $\lim_{x\to-\infty} f(x) = +\infty$ using only inequalities between real numbers. Prove these two statements for $f(x) = 1/x - x^3$.

26. State sequential characterizations of a finite right-hand limit, a finite limit at $+\infty$, and divergence to $+\infty$ at a finite accumulation point. Specify the permitted sequences in each case.

2.3 Properties of limits

Source: §2.3, pp. 16–19.

Uniqueness and boundedness

27. State and prove uniqueness of a finite function limit (Proposition 2.18). Where is the accumulation-point hypothesis used?

28. Give a domain A and a point c that is not an accumulation point of A. Show that without the accumulation hypothesis, the ϵ-δ condition for a limit could hold for every $L \in \mathbb{R}$.

29. Define boundedness of a function on $B \subseteq A$ and boundedness on its domain. Restate the first definition using the image set $f(B)$.

30. Give a function that is unbounded on its domain but bounded on a smaller interval. For $f(x) = 1/x$ on $(0, 1]$, find a bound on $[\delta, 1]$, where $0 < \delta < 1$. For $g(x) = x^2$ on $\mathbb{R}$, find a bound on a finite interval $[a, b]$.

31. State and prove the local boundedness consequence of a finite limit (Proposition 2.21). What bound follows by choosing $\epsilon = 1$? Is boundedness on the entire domain asserted?

32. Does boundedness in a punctured neighborhood imply existence of a limit? Give a counterexample from this chapter.

Algebraic properties

33. State the limit laws for scalar multiples, sums, products, and quotients (Theorem 2.22), including all hypotheses and the additional restriction in the quotient law.

34. Prove the sum law directly from the ϵ-δ definition. How do you choose a radius that works for both functions?

35. Prove the product law directly from the definition. Explain how to split $f(x)g(x) - LM$ into two error terms and where local boundedness is used. Ensure that your choices also work when $L = 0$.

36. If $g(x) \to M \neq 0$, show that $|g(x)|$ is bounded away from zero on the domain sufficiently close to c, with $x \neq c$. How does this ensure that the quotient is defined there? Use an estimate for $|1/g(x) - 1/M|$ to justify the quotient law.

37. Explain how the sequential characterization of function limits transfers the algebraic limit laws for sequences to functions.

Order properties

38. State and prove the order-preservation theorem for finite limits (Theorem 2.23). In the contradiction proof, how do you choose a positive error tolerance from the two proposed limits?

39. If $f(x) < g(x)$ for every domain point near c, must their limits satisfy a strict inequality? Give a counterexample.

40. State the squeeze theorem (Theorem 2.24), including the assumptions that guarantee existence of the middle function's limit.

41. Suppose $|f(x)| \leq g(x)$ near c, with $x \neq c$, and $g(x) \to 0$. What follows about $f(x)$, and how does the squeeze theorem give it?

42. Give an example showing why convergence of the two bounding functions to the same limit is essential in the squeeze theorem. Can boundedness of the middle function alone replace this hypothesis?

3 Continuous Functions

3.1 Continuity

Source: §3.1, pp. 21–25.

1. State the ϵ-δ definition of continuity of $f : A \to \mathbb{R}$ at c. Define continuity on $B \subseteq A$ and continuity of a function on its domain.
2. State the neighborhood definition of continuity at a point. Explain its equivalence with the ϵ-δ definition. How does it differ from the neighborhood definition of a limit?
3. Prove that every function is continuous at each isolated point of its domain. Why is no accumulation-point hypothesis needed in the definition of continuity?
4. At an accumulation point $c \in A$, express continuity in terms of a function limit. What does this imply for a function defined on an open interval?
5. Characterize continuity of $f : [a, b] \to \mathbb{R}$, with $a < b$, using limits at interior points and one-sided limits at the endpoints.
6. Let $A = \{0\} \cup \{1/n : n \in \mathbb{N}\}$, and define $f(0) = y_0$, $f(1/n) = y_n$. At which points is continuity automatic? Give a necessary and sufficient condition on (y_n) and y_0 for f to be continuous on A.
7. State the sequential characterization of continuity (Theorem 3.6). May the testing sequences have terms equal to c? Explain why the characterization also holds if c is isolated.
8. How can a single sequence establish discontinuity at a point? State the conditions that the sequence and its function values must satisfy.
9. Prove directly from the definition that $f(x) = \sqrt{x}$ is continuous on $[0, \infty)$. Give explicit choices of δ, treating $c > 0$ and $c = 0$ separately.
10. Use a trigonometric identity and $|\sin t| \leq |t|$ to prove that sine is continuous at every real point. Can the same choice of δ work at every point for a fixed ϵ?
11. Determine all continuity points of $\operatorname{sgn} x$. What kind of discontinuity does it have at zero, and can changing its value at zero make it continuous there?
12. Define $f(0) = 0$ and $f(x) = 1/x$ for $x \neq 0$. Determine its continuity points and describe its one-sided behavior at zero. Distinguish divergence to infinity from a finite one-sided limit.
13. Define $f(0) = 0$ and $f(x) = \sin(1/x)$ for $x \neq 0$. Determine its continuity points and show why no choice of $f(0)$ can make it continuous at zero.
14. Define $f(0) = a$ and $f(x) = x \sin(1/x)$ for $x \neq 0$. For which $a \in \mathbb{R}$ is f continuous at zero? Prove your answer and classify the discontinuity for the other values.
15. Define the Dirichlet function. Prove that it is discontinuous everywhere and has no limit at any real point. Where do density of the rationals and density of the irrationals enter the argument?
16. Define the version of the Thomae function used in the notes, with value zero at zero and at irrational points, and value $1/q$ at a nonzero rational p/q in lowest terms with $q > 0$. State exactly where it is continuous and where it is discontinuous.
17. Define removable, jump, and essential discontinuities using finite limits, following the terminology in these notes. Give an example of each. Which can be repaired by changing the function's value at the discontinuity?

3.2 Properties of continuous functions

Source: §3.2, pp. 25–26.

18. State the continuity rules for scalar multiples, sums, products, and quotients (Theorem 3.15). Include the extra condition for a quotient and explain how the rules follow from limit laws at accumulation points. What happens at isolated domain points?

19. Define polynomial and rational functions with real coefficients. What is the natural domain of a rational function given by P/Q?

20. State and prove the continuity result for polynomials and rational functions (Corollary 3.16). Which elementary continuous functions and algebraic rules suffice?

21. Determine the domain and continuity points of

$$f(x) = \frac{x + 3x^3 + 5x^5}{1 + x^2 + x^4}.$$

Justify your answer without an ϵ-δ computation.

22. Define $g \circ f$ for $f : A \to \mathbb{R}$ and $g : B \to \mathbb{R}$. What relation between $f(A)$ and B makes the composition well-defined on all of A?

23. State the pointwise and whole-domain continuity theorems for compositions (Theorem 3.18 and Corollary 3.19). At which point must the outer function be continuous when testing continuity at c?

24. Prove continuity of a composition from the definition. Explain the order in which you choose the output tolerance, the intermediate tolerance, and the input radius.

25. Use composition and product rules to justify continuity of $\sin(1/x)$ and $x \sin(1/x)$ at every nonzero point. Why do these rules alone say nothing about continuity at zero?

3.3 Uniform continuity

Source: §3.3, pp. 27–29.

26. Define uniform continuity on A. Compare the quantifier order with continuity at every point of A. Which dependencies of δ are permitted in each definition?

27. Prove that uniform continuity implies continuity at every domain point. Is the converse true?

28. Negate the definition of uniform continuity. State the equivalent sequential criterion for failure of uniform continuity (Proposition 3.23), including the fixed positive separation in function values. Must the two input sequences converge individually?

29. Prove Proposition 3.23 in both directions. How does choosing $\delta = 1/n$ produce the required sequences?

30. Prove that sine is uniformly continuous on $\mathbb{R}$ by giving a single δ for each $\epsilon > 0$.

31. Prove that $x \mapsto x^2$ is uniformly continuous on $[0, 1]$, and then on every bounded subset of $\mathbb{R}$. Give an explicit uniform choice of δ.

32. Show that $x \mapsto x^2$ is not uniformly continuous on $\mathbb{R}$ by constructing two input sequences whose distances tend to zero while their squared values remain separated.

33. Give a continuous function on the bounded interval $(0, 1]$ that is not uniformly continuous. Verify failure using two explicit sequences from that interval.

34. Give a bounded continuous function on $(0, 1]$ that is not uniformly continuous. Construct sequences that establish the failure. Why does boundedness of both the domain and the range not suffice?

3.4 Continuous functions and open sets

Source: §3.4, pp. 29–30.

35. Define the image $f(B)$ and preimage $f^{-1}(C)$ of sets under $f : A \to \mathbb{R}$. Does defining a preimage require f to have an inverse function?

36. For $f(x) = x^2$ on $\mathbb{R}$, compute the image and preimage of $(1, 4)$, and the image and preimage of $(-1, 1)$.

37. Must a continuous function map open sets to open sets? Give a counterexample and distinguish images from preimages.

38. State the characterization of continuity by preimages of open sets (Theorem 3.30). In which space must each preimage be open? Does the characterization concern continuity at one point or throughout the domain?

39. Prove that a continuous $f : A \to \mathbb{R}$ has relatively open preimages of open subsets of $\mathbb{R}$. Construct a relative neighborhood at each point of a preimage.

40. Prove the converse by taking preimages of ϵ-neighborhoods of $f(c)$. How does relative openness provide the required input radius?

3.5 Continuous functions on compact sets

Source: §3.5, pp. 30–32.

41. State the theorem on continuous images of compact sets (Theorem 3.31). Does the image have to be an interval?

42. Prove Theorem 3.31 using sequences. Given a sequence in $f(K)$, how do you obtain a sequence in K, and where do compactness and continuity enter the subsequence argument?

43. Prove Theorem 3.31 using open covers and preimages. The preimages are open in K; how can they be written as intersections with open subsets of $\mathbb{R}$ so that the ambient open-cover definition of compactness applies?

44. Give a continuous function on a closed subset of $\mathbb{R}$ whose image is not closed. Compute the image and identify the missing hypothesis that prevents Theorem 3.31 from applying.

45. State the Weierstrass extreme value theorem (Theorem 3.33), including the nonemptiness assumption needed for attained extrema. Distinguish boundedness from attainment of a supremum and infimum.

46. Deduce the extreme value theorem from compactness of the continuous image. Why do the extrema of the image correspond to values attained at points of the domain?

47. Give an unbounded function on a compact interval. Does it have a minimum or a maximum? Explain which hypothesis of the extreme value theorem fails.

48. Give a continuous, unbounded function on a bounded interval. Does it have a minimum or a maximum? Explain which hypothesis of the extreme value theorem fails.

49. Give a bounded continuous function that attains neither its supremum nor its infimum. Specify the domain and compute both bounds.

50. Define $f : [0, 2/\pi] \to \mathbb{R}$ by $f(0) = 0$ and $f(x) = x + x\sin(1/x)$ for $x > 0$. Show that it is continuous and find its maximum and minimum values. Exhibit infinitely many points where the minimum is attained. Check the printed maximum in Example 3.37 against the value at the right endpoint.

51. State the uniform continuity theorem for continuous functions on compact sets (Theorem 3.38). Which failures of compactness occur in the counterexamples of §3.3?

52. Prove Theorem 3.38 by contradiction using the sequential criterion for failure of uniform continuity. Why do the two selected subsequences converge to the same point, and why does that point belong to the domain?

53. Explain why the oscillatory function in Example 3.37 is uniformly continuous on $[0, 2/\pi]$, despite oscillating infinitely often near zero. Is an explicit uniform radius needed to apply Theorem 3.38?

3.6 The intermediate value theorem

Source: §3.6, pp. 32–35.

54. State the zero-existence version of the intermediate value theorem (Theorem 3.40), including the endpoint sign conditions and the location of the zero.

55. Prove Theorem 3.40 for $f(a) < 0 < f(b)$ using $E = \{x \in [a, b] : f(x) < 0\}$ and $c = \sup E$. Why does c exist? Show why both $f(c) < 0$ and $f(c) > 0$ are impossible, and why the resulting zero lies in (a, b).

56. Give a continuous function on a compact set with positive and negative values but no zero. Explain why the interval hypothesis in Theorem 3.40 matters.

57. Give a function on a compact interval with opposite signs at the endpoints but no zero. Explain why continuity cannot be omitted from Theorem 3.40.

58. Use the intermediate value theorem to prove existence of a positive real solution of $x^2 = 2$. What additional argument gives uniqueness of the positive solution?

59. Describe the bisection method for a continuous function with opposite endpoint signs. Apply the first two bisections to $x^2 - 2$ on $[1, 2]$. How does the enclosing interval's length change after n steps if no zero is found exactly?

60. Restrict $f(x) = x^2 - 2$ to $[1, 2] \cap \mathbb{Q}$. Why is this function continuous on its domain but nowhere zero? What does the example show about the role of real completeness in the intermediate value theorem?

61. State the general intermediate value theorem (Theorem 3.44) and deduce it from the zero-existence version by shifting the function by a constant. How do you handle the two possible orders of the endpoint values?

62. State the result describing the image of a compact interval under a continuous real-valued function (Theorem 3.45). How are the endpoints of the image determined?

63. Prove Theorem 3.45 using the extreme value and intermediate value theorems. Explain separately why the image is contained in the proposed interval and why it contains that entire interval.

64. For $f(x) = x - x^3$ on $[-1, 1]$, compute the image interval using calculus as in Example 3.46. Is it determined by the values at -1 and 1? What extra hypothesis would make $f([a, b]) = [f(a), f(b)]$ valid?

65. Give examples showing how the conclusion of Theorem 3.45 can fail when continuity, the interval hypothesis, or compactness is omitted. In each case, compute the image and specify whether it fails to be an interval, to be closed, or to be bounded.

66. What can be said about the continuous image of an arbitrary interval? Must openness, closedness, or boundedness of the interval be preserved? Use examples from this chapter.

3.7 Monotonic functions

Source: §3.7, pp. 35–37.

67. Define increasing, strictly increasing, decreasing, strictly decreasing, monotonic, and strictly monotonic functions on an interval, using the terminology in these notes.

68. Is a constant function increasing, decreasing, both, or neither under these conventions? Explain the alternative use of "nondecreasing" and "nonincreasing" and why conventions matter.

69. State the one-sided-limit theorem for monotonic functions (Theorem 3.51). At which domain points does it guarantee finite one-sided limits?

70. For an increasing $f : I \to \mathbb{R}$ and an interior point $c \in I$, express its left and right limits as a supremum and an infimum of appropriate sets of values. How do these limits compare with $f(c)$?

71. Prove the left-limit formula in Theorem 3.51 from completeness and monotonicity. Why is the relevant set of values nonempty and bounded above? Explain the corresponding right-limit argument and how to treat decreasing functions.

72. What one-sided limits are guaranteed for a monotonic function on a closed interval $[a, b]$, with $a < b$? Must they agree with the endpoint values? Give an example where they do not.

73. State and prove the restriction on interior discontinuities of monotonic functions (Corollary 3.52). Why can an interior discontinuity be neither removable nor essential?

74. What can be said about the number of discontinuities of a monotonic function? Must it be finite? Contrast this with the discontinuities of the Dirichlet function.

4 Differentiable Functions

4.1 The derivative

Source: §4.1, pp. 39–45.

Definition and examples

1. Define differentiability of $f : (a, b) \to \mathbb{R}$ at c, the derivative $f'(c)$, and differentiability on (a, b). What is the domain of the derivative function?
2. Give the two equivalent difference-quotient definitions of $f'(c)$, using $h \to 0$ and $x \to c$. Why need neither quotient be defined at its limiting input? Explain the secant-slope and tangent-slope interpretation.
3. What does it mean that differentiability is a local property? Explain how the derivative can be defined at any interior point of a domain $A \subseteq \mathbb{R}$, even when A is not an interval.
4. Differentiate $f(x) = x^2$ directly from the definition. Why is cancelling a factor of h legitimate before evaluating the limit, while simply substituting $h = 0$ into the original quotient is not?
5. Let $f(x) = x^2$ for $x > 0$ and $f(x) = 0$ for $x \leq 0$. Find its derivative at every real point, using one-sided limits of the difference quotient at zero.
6. Let $f(0) = 0$ and $f(x) = 1/x$ for $x \neq 0$. Determine where it is differentiable and compute the derivative directly from difference quotients. What happens to the quotient at zero?
7. Determine the differentiability points and derivatives of $\operatorname{sgn} x$ and $|x|$. Compare their continuity and difference quotients at zero.
8. Differentiate $x^{1/3}$ at nonzero points using the difference-of-cubes identity. What happens to the difference quotient at zero? Does an infinite limiting slope count as a derivative in the definition used here?
9. Let $f(0) = 0$ and $f(x) = x \sin(1/x)$ for $x \neq 0$. Determine its differentiability points and its derivative where defined. Prove the conclusion at zero from the definition; the product and chain rules may be used away from zero.
10. Let $f(0) = 0$ and $f(x) = x^2 \sin(1/x)$ for $x \neq 0$. Compute f' at all real points. Is f' continuous at zero? Prove both the existence of $f'(0)$ and your conclusion about continuity of the derivative.

Derivatives as linear approximations

11. Define $r(h) = o(h)$ as $h \to 0$. Is $r(h) \to 0$ alone sufficient to imply $r(h) = o(h)$? Give an example that distinguishes the conditions.
12. State the characterization of differentiability by a linear approximation and a remainder (Proposition 4.10). What is the coefficient of the linear term, and what condition must the remainder satisfy?
13. Prove both directions of Proposition 4.10. How does dividing the error by h connect the linear approximation to the difference-quotient definition?
14. Explain how $f'(c)$ describes the first-order change in $f(c + h) - f(c)$. What do its magnitude and sign say about infinitesimal scaling and orientation? Distinguish the affine approximation to $f(c + h)$ from the linear approximation to its change.

15. For $f(x) = x^2$, and then for $f(x) = 1/x$ at $c \neq 0$, write $f(c+h) = f(c) + f'(c)h + r(h)$ with an exact expression for $r(h)$. Verify the required little-oh condition.

Left and right derivatives

16. Define the right and left derivatives of a function on $[a, b]$, including where each can be considered. Write the left derivative both as a limit with $h \to 0^-$ and as a limit with $h \to 0^+$.
17. Give a necessary and sufficient condition for a two-sided derivative to exist in terms of one-sided derivatives. Illustrate failure of the condition with $|x|$ at zero.
18. Find the endpoint derivatives of $f(x) = x^2$ on $[0, 1]$. Now extend it by setting $f(x) = 1/x$ for $x < 0$ and $f(x) = 0$ for $x > 1$. Which one-sided derivatives at zero and one still exist, and what are their values? Check the right-derivative claim at one in Example 4.14 using the actual value $f(1)$.

4.2 Properties of the derivative

Source: §4.2, pp. 45–49.

Differentiability and continuity

19. State and prove the implication from differentiability to continuity at a point (Theorem 4.16). How can the increment $f(c+h) - f(c)$ be written using the difference quotient?
20. Does continuity at a point imply differentiability there? Give examples from this chapter exhibiting different ways that differentiability can fail for a continuous function.
21. Define continuous differentiability and the notation $f \in C^1(a, b)$. Give an example of a differentiable function that is not continuously differentiable.

Algebraic properties

22. State the scalar-multiple, sum, product, and quotient rules for derivatives (Theorem 4.18), with all hypotheses.
23. Prove the product rule from the difference-quotient definition. Which term do you add and subtract, and where is continuity of a differentiable function needed?
24. Derive the power rule for positive integer powers by induction using the product rule. Deduce differentiability of polynomials and rational functions on their natural domains.

The chain rule

25. State the chain rule (Theorem 4.20), including the domain, interior-point, and differentiability assumptions. At which point is the derivative of the outer function evaluated?
26. Explain the chain-rule formula using composition of the linear approximations to the inner and outer functions.

27. Prove the chain rule using

$$f(c+h) = f(c) + f'(c)h + r(h), \qquad g(f(c)+k) = g(f(c)) + g'(f(c))k + s(k).$$

Identify the full remainder after composition, including the coefficient of $r(h)$. Show that it is $o(h)$. Your argument should also cover $f'(c) = 0$ and increments for which $f(c+h) = f(c)$.

28. Use the chain rule to differentiate $1/f$ where f is differentiable and nonzero. Deduce the quotient rule from the reciprocal formula and the product rule.

The derivative of inverse functions

29. State the formula for the derivative of an inverse function (Proposition 4.22), with every assumption in the proposition. Does this proposition itself establish differentiability of the inverse?

30. Derive the inverse-derivative formula from the identity $f^{-1}(f(x)) = x$. Write the formula at $d = f(c)$ as well as at $f(c)$. Why is differentiability of the inverse impossible there if $f'(c) = 0$ under the stated interior-point assumptions?

31. Apply the inverse-derivative formula to $f(x) = x^3$. Where is its inverse differentiable, and what happens at zero?

32. State the monotonicity and inverse-continuity facts for a continuous one-to-one function on an interval. Under these hypotheses, what extra pointwise conditions guarantee differentiability of the inverse at an interior image point?

33. State the local inverse function theorem quoted in the notes. Specify the differentiability and continuity assumptions near the point and the conclusion about the local inverse and its derivative.

4.3 Extreme values

Source: §4.3, pp. 49–51.

34. Define global and local maxima and minima of $f : A \to \mathbb{R}$. What is an extreme value? Distinguish the point where an extremum is attained from the extreme value itself.

35. State Fermat's necessary condition for a local extremum (Theorem 4.25). Include the assumptions about the point and differentiability of the function.

36. Prove Fermat's theorem using signs of the difference quotient on the two sides of a local maximum. How does the proof change for a local minimum?

37. Why does Fermat's zero-derivative conclusion require an interior point? State the necessary one-sided derivative inequalities at each endpoint of $[a, b]$ for a local maximum and for a local minimum, assuming the relevant derivatives exist.

38. Define critical and stationary points, using the convention in these notes. Must a critical point have a derivative? Are endpoints included in either definition?

39. List the three types of points that must be considered when searching for attained extrema on a domain $A \subseteq \mathbb{R}$. Does being stationary guarantee an extremum? Give a counterexample. Give an example of an interior extremum at a point where the function is not differentiable.

4.4 The mean value theorem

Source: §4.4, pp. 51–53.

40. State Rolle's theorem (Theorem 4.27) on a nondegenerate interval $[a, b]$. Distinguish the continuity requirement at the endpoints from the differentiability requirement in the interior.

41. Prove Rolle's theorem using the extreme value theorem and Fermat's theorem. Handle the constant-function case explicitly. Where does this argument ultimately depend on completeness of $\mathbb{R}$?

42. State the mean value theorem (Theorem 4.28), including every hypothesis and the location of the point it supplies. Give its geometric interpretation.

43. Deduce the mean value theorem from Rolle's theorem. Which affine function should be subtracted from f, and why does the resulting function meet Rolle's hypotheses?

44. Explain how the mean value theorem connects local derivative information to the change between two function values. If $|f'(x)| \leq M$ on (a, b) and f is continuous on $[a, b]$, what bound does it give for $|f(b) - f(a)|$?

45. State and prove the zero-derivative theorem (Theorem 4.29). On what kind of domain is its constancy conclusion asserted? Give an example showing why a domain consisting of two disjoint open intervals can invalidate that conclusion.

46. State and prove the result for two differentiable functions with equal derivatives on an interval (Corollary 4.30). How does the zero-derivative theorem apply?

47. State the derivative-sign characterizations of increasing and decreasing differentiable functions on an interval (Theorem 4.31). State also the sufficient conditions for strict increase and strict decrease.

48. Prove both directions of the increasing-function characterization. Which direction follows from difference quotients, and which uses the mean value theorem? Explain the strict and decreasing versions.

49. Must the derivative of a differentiable strictly increasing function be positive at every point? Give a counterexample and state the derivative inequality that does hold everywhere.

50. Suppose f is differentiable near c, $f'(c) > 0$, and f' is continuous at c. Prove that f is strictly increasing in some neighborhood of c. Where is continuity of the derivative used?

51. Let $f(0) = 0$ and $f(x) = x/2 + x^2 \sin(1/x)$ for $x \neq 0$. Show that $f'(0) > 0$ but f is not increasing on any neighborhood of zero. Find nonzero points arbitrarily close to zero where the derivative is negative.

4.5 Taylor's theorem

Source: §4.5, pp. 53–55.

52. Define the second derivative, higher-order derivatives, and infinite differentiability on an open interval.

53. Define the order-n Taylor polynomial of f at c and its Taylor coefficients. Give the summation formula. Must its actual polynomial degree equal n, or can it be smaller?

54. If P is a polynomial of degree at most n, what is its order-n Taylor polynomial about an arbitrary center c? Verify the claim.

55. Compute the order-n Taylor polynomial of e^x at zero from the definition of the coefficients.

56. Compute the Taylor polynomials of $\cos x$ and $\sin x$ at zero. Specify the nonzero coefficients and explain which successive orders give identical polynomials.

57. Compute the order-n Taylor polynomial of $1/x$ at $c = 1$, deriving the coefficients from its higher derivatives.

58. Compute the order-n Taylor polynomial of $\log x$ at $c = 1$, including its constant term and the general nonzero coefficient.

59. Define the Taylor remainder $R_n(x)$. Why does calculating a Taylor polynomial alone not establish the accuracy of an approximation away from its center?

60. State Taylor's theorem with Lagrange remainder (Theorem 4.41) for a function with $n + 1$ derivatives on an open interval. Specify the location of the intermediate point for $x \neq c$. Does the statement require continuity of the $(n + 1)$st derivative?

61. Show that the case $n = 0$ of Taylor's theorem is the mean value theorem applied between c and x.

62. In the proof of Taylor's theorem, fix $x \neq c$ and define

$$g(t) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x - t)^k.$$

Compute $g(x)$, $g(c)$, and $g'(t)$. Explain the cancellation in the derivative and the relationship between $g(c)$ and the desired remainder.

63. Complete the proof by constructing from $g(t)$ an auxiliary function that vanishes at c and x, as in the notes. Apply Rolle's theorem and derive the Lagrange remainder formula. Why is cancellation of $(x - \xi)^n$ legitimate, and how is the case $x = c$ handled separately?

64. Suppose $|f^{(n+1)}(t)| \leq M$ between c and x. Derive an explicit bound on $|R_n(x)|$. How does the Lagrange remainder differ from simply using the next Taylor coefficient evaluated at c?

65. Use Taylor's theorem to compute

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2}.$$

Give an explicit error bound for the difference between the quotient and its limit, valid for nonzero x. Which Taylor order gives the quadratic error estimate in the notes?

5 Sequences and Series of Functions

5.1 Pointwise convergence

Source: §5.1, pp. 57–59.

1. Define pointwise convergence of $f_n : A \to \mathbb{R}$ to $f : A \to \mathbb{R}$. Write the definition with all quantifiers. On which quantities may the index threshold depend?

2. Find the pointwise limit of $f_n(x) = n/(nx+1)$ on $(0,1)$. Show that each f_n is bounded on this domain and determine whether its limit is bounded. What preservation claim does this disprove?

3. Find the pointwise limit of $f_n(x) = x^n$ on $[0,1]$. Is the limit continuous? What does this show about pointwise limits of continuous functions?

4. On $[0,1]$, define
$$f_n(x) = \begin{cases} 2n^2 x, & 0 \le x \le 1/(2n), \\ 2n^2(1/n - x), & 1/(2n) < x < 1/n, \\ 0, & 1/n \le x \le 1. \end{cases}$$
Find its pointwise limit, its maximum, and the location of that maximum. Why is $(f_n(x))$ bounded for each fixed x, even though there is no common bound over all x and n?

5. Find the pointwise limit of $f_n(x) = \sin(nx)/n$ on $\mathbb{R}$. Does the sequence of derivatives converge pointwise on $\mathbb{R}$? Exhibit a point where it fails and explain what this says about interchanging differentiation and a pointwise limit.

6. Find the pointwise limit on $\mathbb{R}$ of
$$f_n(x) = \frac{x^2}{\sqrt{x^2 + 1/n}}.$$
Explain why every approximating function is differentiable but the limit need not be differentiable everywhere.

7. State the pointwise limit of $(1 + x/n)^n$ on $\mathbb{R}$, using the exponential limit formula quoted in Example 5.7. Does identifying this pointwise limit by itself establish uniform convergence on $\mathbb{R}$?

5.2 Uniform convergence

Source: §5.2, p. 59.

8. Define uniform convergence on A, and compare its quantifier order with pointwise convergence. Which dependence of the threshold N is forbidden in the uniform definition?

9. Prove that uniform convergence implies pointwise convergence to the same limit. Write the negation of uniform convergence to a specified function, including all quantifiers.

10. Prove that x^n does not converge uniformly on $[0,1]$. Does it converge uniformly on $[0,b]$ for $0 \le b < 1$? Give a suitable integer threshold for a specified error tolerance, treating $b = 0$ separately.

11. Show that the triangular functions defined in §5.1 do not converge uniformly to their pointwise limit. Which moving input exposes the failure, and why is checking fixed inputs insufficient?

12. Prove that $\sin(nx)/n$ converges uniformly on $\mathbb{R}$ and give an explicit uniform error bound. Does this stronger convergence fix the failure of derivative convergence found in §5.1?

5.3 Cauchy condition for uniform convergence

Source: §5.3, pp. 60–61.

13. Define a uniformly Cauchy sequence of functions on A. How does this differ from requiring a Cauchy sequence of real values separately at every fixed input?
14. State the uniform Cauchy criterion (Theorem 5.13) and prove that a uniformly convergent sequence is uniformly Cauchy.
15. Prove the converse of Theorem 5.13. First construct the pointwise limit, then show uniform convergence to it. Where is completeness of $\mathbb{R}$ used? If you pass to the limit in a strict Cauchy bound, why might the resulting inequality be non-strict, and how do you still obtain the desired error bound?

5.4 Properties of uniform convergence

Source: §5.4, pp. 61–65.

Boundedness and continuity

16. State and prove the boundedness theorem for uniform limits (Theorem 5.14). Is a common bound for all approximating functions assumed, or only boundedness of each individual function? Why does choosing one sufficiently late function suffice in the proof?
17. If $|f_n(x)| \leq M$ for every n and every $x \in A$, must a pointwise limit be bounded by M? Distinguish this hypothesis from individual boundedness of each f_n.
18. For $f_n(x) = n/(nx+1)$, explain why convergence on $(0,1)$ is not uniform. Prove uniform convergence on $[a,1)$ for $0 < a < 1$, giving an error bound in terms of a and n.
19. State the uniform-limit theorem for continuous functions (Theorem 5.16). Is compactness of the domain a hypothesis?
20. Prove Theorem 5.16 using an $\epsilon/3$ argument. Write the three-term estimate for $|f(x) - f(c)|$. Why do you fix an approximating index before choosing the continuity radius?
21. Express the continuity theorem as an exchange of the limits $n \to \infty$ and $x \to c$, at an accumulation point of the domain. Use x^n at $c = 1$ to show why pointwise convergence alone does not justify this exchange.
22. If continuous functions converge pointwise to a discontinuous function, what follows about uniform convergence? Is continuity of the pointwise limit sufficient for uniform convergence? Give a counterexample from this chapter.

Differentiability

23. For $f_n(x) = x/(1+nx^2)$ on $\mathbb{R}$, find its uniform limit and an explicit uniform error bound. Compute f_n' and its pointwise limit. Does that limit equal the derivative of the limiting function? Is convergence of the derivatives uniform?

24. State the theorem justifying differentiation of a sequence of functions (Theorem 5.18). Which convergence is pointwise and which must be uniform? Are continuity of the derivatives or uniform convergence of the original functions assumed?

25. In the proof of Theorem 5.18, apply the mean value theorem to $f_m - f_n$ between distinct c and x. Use uniform Cauchy bounds for the derivatives and then let $m \to \infty$ to control

$$\left| \frac{f(x) - f(c)}{x - c} - \frac{f_n(x) - f_n(c)}{x - c} \right|.$$

Why is the resulting bound uniform in the choice of $x \neq c$?

26. Complete the proof of Theorem 5.18 by estimating the difference between the difference quotient of f and $g(c)$ with three error terms. Explain the order of the choices of n and the input radius, and why no continuity assumption on g is needed in this proof.

27. If the derivatives in Theorem 5.18 are also continuous, outline the alternative proof using the fundamental theorem of calculus on a compact subinterval. Which passage of a limit through an integral does uniform convergence justify?

5.5 Series

Source: §5.5, pp. 65–66.

28. Define the partial sums of a series of functions, and define pointwise and uniform convergence of the series in terms of them. Is uniform convergence of the individual terms to zero the same as uniform convergence of the series?

29. Compute the finite partial sums of the geometric series $\sum_{k=0}^{\infty} x^k$. Determine all real inputs where it converges, its sum there, and its behavior at the two endpoints.

30. Prove that the geometric series is not uniformly convergent on $(-1, 1)$, but is uniformly convergent on $[-\rho, \rho]$ for every $0 \leq \rho < 1$. Give a uniform bound for the remainder after the nth partial sum.

31. State and prove the uniform Cauchy criterion for series (Theorem 5.21). Which finite tails must be small simultaneously, and how does this follow from the sequence version?

5.6 The Weierstrass M-test

Source: §5.6, pp. 67–69.

32. State the Weierstrass M-test (Theorem 5.22), specifying the domain-independent bounds and the numerical convergence hypothesis. Must the sum of the function series be known in advance?

33. Prove the M-test using the uniform Cauchy criterion and a triangle-inequality estimate on finite tails.

34. Apply the M-test to $\sum_{n=0}^{\infty} x^n$ on $[-\rho, \rho]$, where $0 \leq \rho < 1$. Identify the majorants and compute their sum.

35. Consider
$$f(x) = \sum_{n=1}^{\infty} 2^{-n} \cos(3^n x).$$
Prove uniform convergence on $\mathbb{R}$ and continuity of f. Write the formal term-by-term derivative series. What striking differentiability property of f is reported in Example 5.24, without proof?

36. Define absolute convergence of a series of functions at each point of a domain. Why does the M-test imply absolute convergence as well as uniform convergence? Explain why the two notions of convergence are different.

37. Give a series of constant functions that converges uniformly on $\mathbb{R}$ but is not absolutely convergent at any point. Verify both claims and explain why the M-test cannot apply to this series.

38. Give a series that converges absolutely at every point of its domain but does not converge uniformly there. Why is pointwise absolute convergence not a sufficient replacement for the summable-majorant hypothesis in the M-test?

5.7 The sup-norm

Source: §5.7, pp. 69–70. Assume $A \neq \varnothing$.

39. Define the sup-norm of a function on A. When is it finite? Must the supremum be attained, and is the expression a finite-valued norm on all real-valued functions?

40. Compute the sup-norms on $(0, 1)$ of x^2, $x^2 - x$, and $1/x$. For each finite supremum, decide whether it is attained.

41. Characterize uniform convergence using $\|f_n - f\|$. Explain why this is equivalent to the quantified definition, even though a pointwise strict bound may give only a non-strict bound after taking a supremum.

42. Formulate the uniform Cauchy condition using the sup-norm. How does the notation parallel the Cauchy condition for real numbers?

5.8 Spaces of continuous functions

Source: §5.8, pp. 70–72. Assume K is nonempty and compact.

43. Define $C(K)$. State its vector-space operations and explain why they preserve membership in $C(K)$. What are its zero vector and additive inverses?

44. Define a norm on a real vector space and a normed vector space. State all norm axioms, including finiteness, and explain how a norm determines a distance between vectors.

45. Prove that the sup-norm makes $C(K)$ a normed vector space (Theorem 5.33). Where is compactness needed? In the homogeneity identity, what happens when the scalar is negative?

46. Define convergence and the Cauchy property in a normed vector space. What do they mean specifically for $C(K)$ equipped with the sup-norm?

47. Define completeness of a normed vector space and a Banach space. Must the limit of a Cauchy sequence belong to the same space?

48. State and prove completeness of $C(K)$ in the sup-norm (Theorem 5.36). Use both the uniform Cauchy criterion and the continuity theorem for uniform limits. Why is existence of a real-valued pointwise limit alone insufficient for this conclusion?

6 Power Series

6.1 Introduction

Source: §6.1, pp. 73–74.

1. Define a power series centered at c, its coefficients, and its partial sums. When is it a polynomial? Why does writing down a formal series not yet define a function at every real input?
2. Give a power series centered at zero and one centered at one, identifying their coefficients. Explain how a change of variable reduces a convergence question at a general center to one at zero.

6.2 Radius of convergence

Source: §6.2, pp. 74–76.

3. State the convergence theorem for power series (Theorem 6.2). Specify where convergence is absolute, where divergence occurs, where uniform convergence is guaranteed, and where the sum is continuous. What does the theorem leave undecided at a finite positive radius?
4. Suppose $\sum_{n=0}^{\infty} a_n x_0^n$ converges at $x_0 \neq 0$. Prove absolute convergence for every $|x| < |x_0|$ by bounding the terms at x_0 and comparing with a geometric series.
5. Define the radius at center zero as a supremum of distances where the series converges. Explain why this gives the asserted convergence and divergence regions. Treat radii zero and infinity explicitly, including convergence at the center.
6. For $0 \leq \rho < R$, prove uniform convergence on $|x - c| \leq \rho$ by choosing $\rho < \sigma < R$ and applying the M-test. How does continuity of the sum follow without assuming uniform convergence on the entire open convergence interval?
7. Define radius of convergence and interval of convergence. For $0 < R < \infty$, list the possible endpoint patterns. What are the convergence sets when $R = 0$ and $R = \infty$?
8. State the coefficient-ratio formula for a radius of convergence (Theorem 6.4), including the eventual nonzero-coefficient and limit assumptions. Give the absolute-value formula and its interpretation when the ratio tends to zero or infinity.
9. Derive Theorem 6.4 from the numerical ratio test. How does the ratio of successive series terms depend on $|x - c|$? Why must the center be handled separately?
10. State Hadamard's formula for the radius in terms of $\limsup_{n\to\infty} |a_n|^{1/n}$ (Theorem 6.5), with its zero and infinite cases. Why does this formula apply even when the coefficient-ratio formula is unavailable?
11. Derive Hadamard's formula from the root test. Explain why neither the root-test threshold nor the ratio-test threshold alone determines convergence at an endpoint.

6.3 Examples of power series

Source: §6.3, pp. 76–79.

12. Find the radius, interval, and sum of $\sum_{n=0}^{\infty} x^n$. Compare the convergence set with the domain of the rational function that gives the sum. Is convergence uniform on the entire convergence interval?

13. Determine the radius and interval of convergence of $\sum_{n=1}^{\infty} x^n/n$, testing both endpoints separately and distinguishing absolute from conditional convergence. Does the series converge uniformly on $(-1, 1)$?

14. Find the radius and interval of convergence of $\sum_{n=0}^{\infty} x^n/n!$. What familiar function will its sum define? What uniform convergence on bounded intervals follows from the radius theorem?

15. Find the radii and intervals of convergence of

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \qquad \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}.$$

Identify their sums. How can a ratio test on the displayed terms be used even though the full sequences of power-series coefficients contain zeros?

16. Find the radius and convergence set of $\sum_{n=0}^{\infty} n! x^n$. For a fixed $x \neq 0$, explain why the terms eventually grow in magnitude rather than tending to zero.

17. Determine the center, radius, and interval of convergence of

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (x-1)^n.$$

Compute the coefficient ratio carefully and test each endpoint, including the signs of its terms.

18. Write the lacunary series $\sum_{k=0}^{\infty} (-1)^k x^{2^k}$ in the form $\sum_{n=0}^{\infty} a_n x^n$ by specifying every coefficient. Find its radius and interval of convergence using comparison and then Hadamard's formula. Why is the usual coefficient-ratio formula unsuitable?

19. What is meant by a lacunary power series? For the preceding example, distinguish convergence of the series at $x = 1$ from existence of the limit of its sum as $x \to 1^-$. What behavior of the sum near one is reported in the notes?

6.4 Differentiation of power series

Source: §6.4, pp. 79–82.

20. State the radius-preservation theorem for the formal term-by-term derivative of a power series (Theorem 6.14). Does it assert that the two series have the same endpoint behavior?

21. Prove absolute convergence of the differentiated series at $|x - c| < R$ by choosing $|x - c| < \rho < R$. Explain why the factors nr^{n-1}, for $0 < r < 1$, remain bounded and how this controls the extra factor of n. Handle $x = c$ separately.

22. Prove the reverse radius inequality in Theorem 6.14: if the differentiated series converges at distance $d > 0$ from its center, show that the original series converges absolutely at every smaller distance. Why would divergence of an absolute-value series alone be insufficient to exclude conditional convergence of its signed series?

23. State and prove the theorem permitting differentiation of a convergent power series inside its radius (Theorem 6.15). Apply the differentiation theorem for function sequences to partial sums on smaller intervals. Why is uniform convergence on the entire open convergence interval unnecessary?

24. State the smoothness and coefficient-recovery theorem (Theorem 6.16). Derive the general formula for the kth derivative series and evaluate it at the center to recover each a_k. What radius do all these derivative series have?

25. State and prove uniqueness of power-series coefficients when two convergent power series with the same center agree on a neighborhood of that center (Corollary 6.17). Why is a positive radius of convergence important?

6.5 The exponential function

Source: §6.5, pp. 82–84.

26. Define $E(x)$ by the exponential power series, without presupposing properties of e^x. Prove that E is smooth on $\mathbb{R}$, that $E' = E$, and that $E(0) = 1$.

27. State the addition law for E (Proposition 6.18). What rearrangement of the product of two power series appears in its proof, and why is that rearrangement justified?

28. Prove the addition law by collecting terms of total degree n in the product $E(x)E(y)$ and using the binomial theorem. Why would assuming the usual exponential addition law make this proof circular?

29. Derive the reciprocal relation between $E(-x)$ and $E(x)$. Prove that $E(x) > 0$ for every real x, including negative inputs.

30. State and prove that $E(x)$ grows faster than every nonnegative integer power of x as $x \to \infty$ (Proposition 6.19). Which single positive term of its power series gives a useful lower bound? Extend the limit to any polynomial divided by $E(x)$.

31. State the initial-value characterization of the exponential (Proposition 6.20). Which differential equation and initial condition determine it uniquely on $\mathbb{R}$?

32. Prove Proposition 6.20 by differentiating f/E. Explain why the quotient is defined everywhere, why its derivative vanishes, and how the initial condition fixes the constant.

33. How can the real logarithm be defined from E? Use positivity, the derivative, the addition law, growth, and continuity to justify that E is a bijection from $\mathbb{R}$ onto $(0, \infty)$.

6.6 Taylor's theorem and power series

Source: §6.6, pp. 84–89.

Smooth functions and analytic functions

34. Distinguish the finite-order statement of Taylor's theorem from representation of a function by an infinite power series. Which limiting variable changes when the Taylor order is fixed, and which changes when the input is fixed?

35. Define $C^k(a, b)$ and $C^\infty(a, b)$, including the meaning of smoothness. Why does existence of derivatives of every order force all of them to be continuous, while existence of only k derivatives need not make the kth derivative continuous?

36. Define a real analytic function on an interval. What is required at every center, and must one power series represent the function throughout the entire interval?

37. Prove that every analytic function is smooth, and explain why its local power-series coefficients must be its Taylor coefficients. Does the converse follow from this argument?

38. For a smooth function, define its formal Taylor series at a point. Describe the two possible failures discussed in the notes that prevent this formal series from representing the function in a neighborhood of the point.

A smooth, non-analytic function

39. Define
$$\phi(x) = \begin{cases} \exp(-1/x), & x > 0, \\ 0, & x \le 0. \end{cases}$$
Prove continuity at zero and compute $\phi'(0)$ from both sides. How does exponential growth at infinity control the right-hand difference quotient?

40. Show inductively that, for $x > 0$, $\phi^{(n)}(x) = p_n(1/x)\exp(-1/x)$ for a polynomial p_n. Use this form to prove existence of every derivative at zero and determine all the values $\phi^{(n)}(0)$.

41. Find the Taylor series of ϕ at zero, its radius of convergence, and its sum. Why does this establish that a smooth function need not be analytic, even when its Taylor series converges everywhere?

42. Why do the vanishing Taylor polynomials of ϕ not contradict Taylor's theorem? Explain how bounds of the form $0 \le \phi(x) \le C_{n+1}x^{n+1}$, with constants depending on n, describe behavior as $x \to 0^+$ without forcing $\phi(x) = 0$ at every fixed positive input.

43. Give a smooth function that is positive for every nonzero real input and has all derivatives zero at zero. Construct it by composing ϕ with x^2, and explain its smoothness and failure of analyticity at zero.

44. Define compact support for a function on $\mathbb{R}$. Give a nonzero continuous function with compact support and explain why a piecewise-linear triangular example is not smooth.

45. Construct a nonzero smooth function with compact support using $\phi(1 - x^2)$. Write its piecewise formula, determine where it is positive and where it vanishes, and justify smoothness at the two transition points.

46. Compare what the value and all derivatives at one point determine for smooth functions and for analytic functions on an interval, using the uniqueness property quoted in the notes. Why can a nonzero analytic function on $\mathbb{R}$ not have compact support? How does this apply to the smooth bump function above?

6.7 Appendix: Review of series

Source: §6.7, pp. 89–92.

47. Define convergence and absolute convergence of a numerical series using partial sums. Distinguish a term, a partial sum, and the sum of the series.

48. State and prove the Cauchy criterion for a real series (Theorem 6.29). Specify which finite tails must satisfy the bound and explain the role of completeness of $\mathbb{R}$.

49. Prove that an absolutely convergent real series converges, using the Cauchy criterion and the triangle inequality. Give a convergent series that is not absolutely convergent.

50. State and prove the necessary condition on the terms of a convergent series (Theorem 6.30). Is the condition sufficient? Give a counterexample to any invalid converse.

51. State and prove the comparison test (Theorem 6.31), including its sign and absolute-value conditions. Which series is shown to converge absolutely?

52. State the ratio test (Theorem 6.32), including eventual nonvanishing of terms, existence of the absolute ratio limit, and the case where it diverges to infinity. What happens below, above, and at the threshold one?

53. Prove the two decisive cases of the ratio test by comparison with geometric behavior. When the ratio limit is less than one, how is a geometric upper bound obtained? When it is greater than one, why do the terms fail to approach zero?

54. Define the limit superior of a real sequence by tail suprema. Why is the tail-supremum sequence decreasing? Explain the conventions giving finite, $+\infty$, and $-\infty$ values. Can a limit superior exist when the original sequence has no limit?

55. State the eventual-upper-bound and infinitely-often-lower-bound characterization of a finite limit superior (Proposition 6.34). State also the versions for $+\infty$ and $-\infty$. Distinguish an eventual condition from one holding infinitely often.

56. State the root test (Theorem 6.35) using $\limsup_{n\to\infty} |a_n|^{1/n}$. Give the conclusions when this number is below one or above one, including the zero and infinite cases. What does the test say when it equals one?

57. Prove the convergent case of the root test by choosing a number strictly between the limit superior and one. How does Proposition 6.34 yield a geometric bound on all sufficiently late terms?

58. Prove the divergent case of the root test using a number strictly between one and the limit superior. Why are infinitely many large terms enough to establish divergence, even without a lower bound on every late term?

59. Compare the hypotheses of the ratio and root tests. Which accommodates zero terms and oscillating root magnitudes? Use the lacunary power series of §6.3 to illustrate why the more general test is useful.